

SPRING AND WEDGE CONSTRAINTS

JEE Advanced Preparation — Physics

Topic 03 | Wedge Motion, Normal Forces, Pseudo Force, Spring-Wedge, Energy Method

Configuration	Wedge Accel A	Key Result
Free smooth wedge + smooth block	$mg\sin\theta\cos\theta / (M+m\sin^2\theta)$	Momentum conserved horiz.
Fixed wedge, smooth block	0 (fixed)	$a_{\text{block}} = g\sin\theta$
Fixed wedge, rough block	0 (fixed)	$a = g(\sin\theta - \mu\cos\theta)$
Block in vertical guides	$(F - mg\tan\theta) / (M + m\tan^2\theta)$	$a_{\text{block}} = A\tan\theta$
Block stationary on smooth wedge	$A = g\tan\theta$	Critical accel condition
Spring-wedge (horiz. spring)	Equilibrium: $A=0$	$kx = mg\tan\theta$

Wedge-block problems are among the most challenging in JEE Advanced. They test the student's ability to correctly identify constraint equations from geometry, draw accurate FBDs, and solve coupled Newton's law equations. The pseudo-force method provides a powerful alternative approach.

KEY FORMULAS

Free smooth wedge: $A = \frac{m g \sin\theta \cos\theta}{M + m \sin^2\theta}$

Block acceleration along slope: $a_{\text{rel}} = \frac{(M+m) g \sin\theta}{M + m \sin^2\theta}$

Normal force: $N = \frac{M m g \cos\theta}{M + m \sin^2\theta}$

Pseudo force in wedge frame: $F_p = m A$ (horizontal, opposite to wedge accel direction)

Block stationary on smooth wedge: $A_{\text{critical}} = g \tan\theta$

Vertical block on wedge: $a_{\text{block}}(\text{vertical}) = a_{\text{wedge}}(\text{horizontal}) \cdot \tan\theta$

Spring equilibrium (horizontal spring on fixed wedge): $kx = m g \tan\theta$

Momentum conservation: $M V_{\text{wedge}} + m v_{x \text{ block}} = \text{const}$ (no external horizontal force)

SECTION 1: WEDGE GEOMETRY AND FBD ANALYSIS

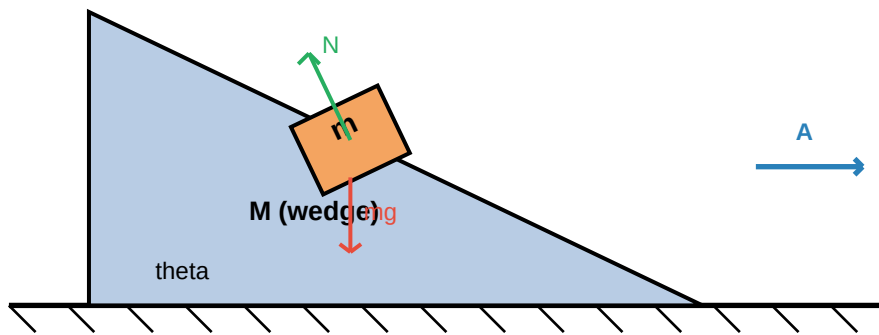


Fig 1: Free smooth wedge on smooth floor with block on incline

$$A = \frac{mg \sin \theta \cos \theta}{M + m \sin^2 \theta}$$

$$N = \frac{Mmg \cos \theta}{M + m \sin^2 \theta}$$

Constraint derivation for free wedge:

Let wedge move right by X (velocity V , acceleration A).

Block moves along wedge surface: displacement s along slope, velocity v_{rel} relative to wedge.

Block's ground-frame velocity: $v_x = V - v_{\text{rel}} \cos(\theta)$; $v_y = -v_{\text{rel}} \sin(\theta)$

FBD — Wedge (horizontal Newton's law):

$N \sin(\theta) = M A$... (i) [N = normal from block, directed into wedge perpendicularly]

FBD — Block (Newton's laws):

Along slope (down+): $m g \sin(\theta) = m a_{\text{rel}} - m A \cos(\theta)$... (ii)

Perp to slope: $N - m g \cos(\theta) = m A \sin(\theta)$... (iii)

Solving:

From (iii): $N = m g \cos(\theta) + m A \sin(\theta)$

Substituting in (i): $m \sin(\theta) [g \cos(\theta) + A \sin(\theta)] = M A$

$\Rightarrow A (M + m \sin^2(\theta)) = m g \sin(\theta) \cos(\theta)$

$\Rightarrow A = \frac{m g \sin(\theta) \cos(\theta)}{M + m \sin^2(\theta)}$

$\Rightarrow a_{\text{rel}} = g \sin(\theta) + A \cos(\theta)$ [from ii] = $\frac{(M+m) g \sin(\theta)}{M + m \sin^2(\theta)}$

Limiting cases:

$M \rightarrow \infty$ (fixed): $A \rightarrow 0$, $a_{\text{rel}} \rightarrow g \sin(\theta)$. Block slides freely on fixed incline. Correct!

$\theta \rightarrow 0$ (flat): $A \rightarrow 0$, $a_{\text{rel}} \rightarrow 0$. No motion on flat surface. Correct!

$M = 0$ (massless wedge): $A \rightarrow g \cos(\theta) / \sin(\theta)$... complex, but block drops vertically.

IMPORTANT: The denominator is $(M + m \sin^2 \theta)$, not $(M + m)$. This is because the block's inertia contributes through its mass times the sine-squared of the angle. Only the horizontal component of the block's acceleration is directly linked to the wedge acceleration.

SECTION 2: PSEUDO FORCE METHOD

In the non-inertial frame of the wedge (accelerating at A rightward):
Add pseudo force on block: $F_p = m \cdot A$ (leftward, horizontal).

Block forces in wedge frame:

1. mg downward 2. N perpendicular to slope 3. mA leftward (pseudo)

Along slope (down+):

$$mg \sin(\theta) - mA \cos(\theta) = m a_{\text{rel}}$$

$$\Rightarrow a_{\text{rel}} = g \sin(\theta) - A \cos(\theta)$$

Perpendicular to slope:

$$N = mg \cos(\theta) + mA \sin(\theta)$$

Critical condition for block stationary on wedge ($a_{\text{rel}} = 0$):

$$g \sin(\theta) = A \cos(\theta) \Rightarrow A = g \tan(\theta)$$

At this wedge acceleration, the pseudo force exactly cancels the gravitational component along slope.

Effective gravity in accelerating frame:

The combination of g (downward) + A (leftward pseudo) gives effective g_{eff} .

$$g_{\text{eff}} = \sqrt{g^2 + A^2}, \text{ tilted at angle } \phi = \arctan(A/g) \text{ from vertical.}$$

All formulas in this frame use g_{eff} and the angle $(\theta - \phi)$ for the slope.

This is the most elegant way to solve problems where the wedge has a known acceleration.

SECTION 3: VERTICAL BLOCK CONSTRAINT

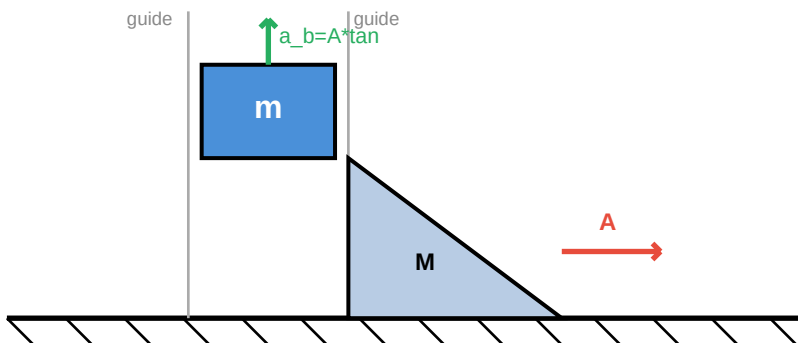


Fig 2: Vertical block constraint

Geometric constraint (fundamental):

If wedge moves right by dx , block rises by $dx \cdot \tan(\theta)$. [θ from vertical]

$$v_{\text{block(up)}} = v_{\text{wedge(right)}} \cdot \tan(\theta)$$

$$a_{\text{block(up)}} = a_{\text{wedge(right)}} \cdot \tan(\theta)$$

Force analysis:

Normal N between block and wedge (perpendicular to wedge face at angle θ from vertical).

$N \cdot \cos(\theta)$ acts vertically (upward on block); $N \cdot \sin(\theta)$ acts horizontally (pushes wedge back).

Block (vertical): $N \cdot \cos(\theta) - mg = m \cdot a_{\text{block}} = m \cdot a_{\text{w}} \cdot \tan(\theta) \dots (1)$

Wedge (horizontal): $F - N \cdot \sin(\theta) = M \cdot a_{\text{w}} \dots (2)$

From (1): $N = m \cdot (g + a_{\text{w}} \cdot \tan(\theta)) / \cos(\theta)$

Substitute in (2): $F - m \cdot \tan(\theta) \cdot (g + a_{\text{w}} \cdot \tan(\theta)) = M \cdot a_{\text{w}}$

$$F - mg \cdot \tan(\theta) = a_{\text{w}} \cdot (M + m \cdot \tan^2(\theta))$$

$$a_{\text{w}} = (F - mg \cdot \tan(\theta)) / (M + m \cdot \tan^2(\theta))$$

$$a_{\text{block}} = a_{\text{w}} \cdot \tan(\theta) = (F - mg \cdot \tan(\theta)) \cdot \tan(\theta) / (M + m \cdot \tan^2(\theta))$$

If no external force $F = 0$ (block descends due to gravity, wedge moved by reaction):

$$a_{\text{w}} = -mg \cdot \tan(\theta) / (M + m \cdot \tan^2(\theta)) \text{ [negative means wedge moves left, if block descends]}$$

$$a_{\text{block}} = -mg \cdot \tan^2(\theta) / (M + m \cdot \tan^2(\theta)) \text{ [downward for block]}$$

SECTION 4: SPRING-WEDGE PROBLEMS — DETAILED ANALYSIS

A. Spring along slope, block on fixed wedge:

Spring constant k , along the slope, connects block to top of wedge (fixed).

Equilibrium: $kx = mg\sin(\theta) \Rightarrow x = mg\sin(\theta)/k$

Oscillation: $\omega = \sqrt{k/m}$, $T = 2\pi\sqrt{m/k}$ (spring along slope = effective spring constant k)

B. Horizontal spring, block on fixed wedge:

Spring connects block to a vertical wall. Block can slide on smooth incline.

Equilibrium: resolve forces along slope.

$kx\cos(\theta) = mg\sin(\theta)$ [spring force has component $kx\cos(\theta)$ up the slope]

$kx = mg\tan(\theta) \Rightarrow x = mg\tan(\theta)/k$

Effective spring constant for oscillation along slope: $k_{\text{eff}} = k\cos^2(\theta)$

$T = 2\pi\sqrt{m / (k\cos^2(\theta))}$

C. Spring (horizontal) connects wedge to wall, block on free wedge:

Equilibrium: block in equilibrium on smooth wedge requires $A = g\tan(\theta)$.

Spring provides the force to maintain this: $kx_0 = M A = M g\tan(\theta)$.

$x_0 = M g\tan(\theta)/k$ (equilibrium spring compression/extension).

Dynamic: perturb from equilibrium, system oscillates. The full oscillation involves coupled motion of wedge and block; reduced mass analysis needed.

General principle for spring-wedge:

1. Identify degrees of freedom (how many independent coordinates).
2. Write energy T (kinetic) and V (potential) in terms of these coordinates.
3. Use Lagrange equations or Newton's laws directly.
4. Find equilibrium positions, then linearize for small oscillations.

SECTION 5: MOMENTUM AND ENERGY METHODS

Momentum Conservation (horizontal):

For smooth wedge on smooth floor: no external horizontal force.

$$d/dt(P_{\text{total_horizontal}}) = 0$$

$$M*V + m*v_x = \text{const} = 0 \text{ (if released from rest)}$$

$$\Rightarrow V = -m*v_x/M \text{ (wedge moves opposite to block's horizontal motion)}$$

Energy Conservation:

Total mechanical energy = KE_wedge + KE_block + PE_gravity (+ PE_spring if present).

$$(1/2)*M*V^2 + (1/2)*m*(v_x^2 + v_y^2) = m*g*h \text{ (if block descends } h, \text{ no spring)}$$

Geometric constraint on velocities:

Block slides along wedge surface. Relative velocity of block w.r.t. wedge must be along slope.

$$v_x(\text{block}) - V = v_{\text{rel}}*(-\cos(\theta)) \text{ [block moves left relative to wedge as it slides down]}$$

$$v_y(\text{block}) = -v_{\text{rel}}*\sin(\theta) \text{ [block moves down as it slides down slope]}$$

$$\text{Eliminating } v_{\text{rel}}: v_y(\text{block}) / (V - v_x(\text{block})) = \tan(\theta)$$

Combined with $V = -m*v_x/M$: this uniquely determines all velocities.

Example calculation:

Wedge $M=4$ kg, block $m=1$ kg, $\theta=45$ deg. Released from rest. Block descends $h=0.2$ m.

$$\text{Momentum: } V = -v_x/4 \text{ (since } m/M = 1/4)$$

$$\text{Geometry: } v_y/(V-v_x) = \tan 45 = 1 \Rightarrow v_y = V - v_x = -v_x/4 - v_x = -5v_x/4$$

$$\text{Energy: } 1*10*0.2 = 0.5*4*(v_x/4)^2 + 0.5*1*(v_x^2 + (5v_x/4)^2)$$

$$2 = 0.5*4*(v_x^2/16) + 0.5*(v_x^2 + 25v_x^2/16)$$

$$2 = v_x^2/8 + v_x^2/2 + 25v_x^2/32 = v_x^2*(4/32+16/32+25/32) = v_x^2*45/32$$

$$v_x^2 = 64/45. \quad v_x = 1.19 \text{ m/s. } V = 0.298 \text{ m/s (wedge). Speed of block} = \sqrt{v_x^2 + v_y^2}.$$

WORKED EXAMPLES — JEE ADVANCED LEVEL

Example 1: Free Smooth Wedge Released from Rest

Problem: Wedge $M=3$ kg, $\theta=30^\circ$, block $m=1$ kg. All smooth. Find A and a_{rel} .

$$A = m g \sin \theta \cos \theta / (M + m \sin^2 \theta) = 1 \cdot 10 \cdot 0.5 \cdot 0.866 / (3 + 0.25) = 4.33 / 3.25 = 1.33 \text{ m/s}^2$$

$$a_{\text{rel}} = (M + m) g \sin \theta / (M + m \sin^2 \theta) = 4 \cdot 10 \cdot 0.5 / 3.25 = 6.15 \text{ m/s}^2$$

$$N = m (g \cos \theta + A \sin \theta) = 1 (8.66 + 0.665) = 9.33 \text{ N}$$

Block in ground frame: $a_x = A - a_{\text{rel}} \cos 30 = 1.33 - 5.33 = -4 \text{ m/s}^2$; $a_y = -a_{\text{rel}} \sin 30 = -3.08 \text{ m/s}^2$

Example 2: Spring Equilibrium on Fixed Wedge (Horizontal Spring)

Problem: Fixed wedge $\theta=45^\circ$, block $m=2$ kg, horizontal spring $k=200$ N/m. Equilibrium extension?

$$\text{Along slope: } kx \cos 45 = mg \sin 45 \Rightarrow kx = mg \Rightarrow x = mg/k = 2 \cdot 10 / 200 = 0.10 \text{ m}$$

$$\text{Oscillation } k_{\text{eff}} = k \cos^2(45) = 200 \cdot 0.5 = 100 \text{ N/m}$$

$$T = 2\pi \sqrt{m/k_{\text{eff}}} = 2\pi \sqrt{2/100} = 0.889 \text{ s}$$

$x = 10$ cm equilibrium extension.

Example 3: Rough Fixed Wedge

Problem: Block m on fixed wedge $\theta=30^\circ$, $\mu_k=0.3$. Block slides. Find acceleration.

$$a = g(\sin 30 - \mu_k \cos 30) = 10(0.5 - 0.3 \cdot 0.866) = 10(0.5 - 0.26) = 2.4 \text{ m/s}^2$$

For equilibrium need $\mu_s \geq \tan 30 = 0.577$. Since $\mu_k=0.3 < 0.577$, block slides.

Example 4: Vertical Block Constraint with External Force

Problem: Block $m=2$ kg in vertical guides. Smooth wedge $M=5$ kg, $\theta=45^\circ$. Force $F=50$ N on wedge. a_{block} ?

$$\text{Constraint: } a_{\text{block}} = a_{\text{wedge}} (\tan 45 = 1).$$

$$\text{Block (vertical): } N/\sqrt{2} - 20 = 2a_w \dots (1). \text{ Wedge (horizontal): } 50 - N/\sqrt{2} = 5a_w \dots (2)$$

$$(1)+(2): 30 = 7a_w \Rightarrow a_w = 30/7 = 4.29 \text{ m/s}^2. a_{\text{block}} = 4.29 \text{ m/s}^2 \text{ upward.}$$

$$N = \sqrt{2}(20 + 2 \cdot 4.29) = \sqrt{2} \cdot 28.57 = 40.4 \text{ N}$$

Example 5: Critical Acceleration for Block Stationary

Problem: Smooth wedge $\theta=30^\circ$, pushed horizontally. Find A for block to not slide.

$$A_{\text{critical}} = g \tan(30) = 10/\sqrt{3} = 5.77 \text{ m/s}^2$$

At this A : pseudo force mA leftward; component along slope $= mA \cos 30 = m \cdot 5.77 \cdot 0.866 = m \cdot 5 \text{ N}$

Gravity along slope $= mg \sin 30 = m \cdot 5 \text{ N}$. They balance: net along-slope force $= 0$.

$A = g \tan(\theta)$ is the universal condition for smooth wedge.

Example 6: Energy and Momentum Method

Problem: Free smooth wedge $M=4$ kg, $\theta=45^\circ$, block $m=1$ kg. Block descends $h=0.2$ m. Find wedge speed.

Momentum: $4V + 1v_x = 0 \Rightarrow v_x = -4V$

Geometry: $v_y/(V-v_x)=\tan 45=1 \Rightarrow v_y = V-(-4V) = 5V$ (downward)

Energy: $mgh = \frac{1}{2}Mv^2 + \frac{1}{2}m(v_x^2+v_y^2)$

$$1 \cdot 10 \cdot 0.2 = 2V^2 + 0.5(16V^2 + 25V^2) = 2V^2 + 20.5V^2 = 22.5V^2$$

$$V^2 = 2/22.5 = 0.0889. \quad V = 0.298 \text{ m/s.}$$

Block speed: $\sqrt{16 \cdot 0.0889 + 25 \cdot 0.0889} = \sqrt{41 \cdot 0.0889} = \sqrt{3.64} = 1.91 \text{ m/s}$

PRACTICE QUESTIONS — JEE ADVANCED LEVEL

Q1. Block on smooth wedge θ , wedge on smooth floor, $A = g \tan(\theta)$. Block relative to wedge:

(A) slides down (B) slides up (C) stationary (D) oscillates

Q2. Normal force N between m and free smooth wedge M (angle θ):

(A) $mg \cos(\theta)$ (B) $Mmg \cos / (M + m \sin^2)$ (C) $(M + m)g \cos$ (D) mg

Q3. Smooth wedge $\theta = 45^\circ$ moves at $v = 2$ m/s horizontally; block in vertical guides. Block speed:

(A) 2 m/s (B) 4 m/s (C) $\sqrt{2}$ m/s (D) 1 m/s

Q4. Fixed wedge $\theta = 45^\circ$, horizontal spring $k = 100$, mass $m = 1$ kg. Equilibrium spring stretch x :

(A) 0.1 m (B) 0.05 m (C) 0.2 m (D) 0.14 m

Q5. Fixed wedge $\theta = 30^\circ$, $\mu_k = 0.4$. Block acceleration along slope ($g = 10$):

(A) 1.54 m/s^2 (B) 5 m/s^2 (C) 2.4 m/s^2 (D) 3.0 m/s^2

Q6. Free smooth wedge + smooth block released. Horizontal momentum of system is:

(A) Increases (B) Decreases (C) Zero always (D) Constant (= initial value)

Q7. Smooth block on smooth free wedge $\theta = 30^\circ$. Wedge accel $A = 5 \text{ m/s}^2$ (given). Relative accel of block:

(A) $g \sin 30 - A \cos 30$ (B) $g \sin 30 + A \cos 30$ (C) $A \tan 30$ (D) $g \cos 30$

Q8. For wedge $M = 3$ kg, $\theta = 30^\circ$, block $m = 1$ kg, smooth: Normal force $N =$

(A) $9.33 \text{ N/kg} \cdot m$ (B) $mg \cos 30$ (C) $Mg \cos 30$ (D) $(m + M)g$

Q9. Block constrained vertically, wedge pushed right. If wedge accelerates at A , block's vertical accel:

(A) A (B) $A \sin(\theta)$ (C) $A \tan(\theta)$ (D) $A \cos(\theta)$

Q10. In spring-wedge system (horizontal spring, fixed wedge θ), frequency of block oscillation:

(A) $\sqrt{k/m} / (2\pi)$ (B) $\cos(\theta) \sqrt{k/m} / (2\pi)$ (C) $\sqrt{k \cos^2 \theta / m} / (2\pi)$ (D) $\sqrt{k \sin^2 \theta / m} / (2\pi)$

ANSWER KEY AND EXPLANATIONS

Q1 — (C) stationary

$A = g \tan(\theta)$ exactly cancels gravity component along slope. No relative motion.

Q2 — (B) $Mmg \cos / (M + m \sin^2)$

Derived from simultaneous Newton's law solution for free wedge-block system.

Q3 — (A) 2 m/s

$v_{\text{block}} = v_{\text{wedge}} \tan(45) = 2 \cdot 1 = 2 \text{ m/s}$. (θ is from vertical here = 45 deg.)

Q4 — (A) 0.1 m

$kx \cos 45 = mg \sin 45 \Rightarrow kx = mg \Rightarrow x = 1 \cdot 10 / 100 = 0.1 \text{ m}$.

Q5 — (A) 1.54 m/s²

$a = g(\sin 30 - 0.4 \cos 30) = 10(0.5 - 0.346) = 10 \cdot 0.154 = 1.54 \text{ m/s}^2$.

Q6 — (D) Constant (= initial value = 0)

No external horizontal force; p_x is conserved. If started from rest, $p_x = 0$ always.

Q7 — (A) $g \sin 30 - A \cos 30$

In pseudo force frame: $a_{\text{rel}} = g \sin(\theta) - A \cos(\theta)$.

Q8 — (A) 9.33 N/kg * m

$N = m(g \cos 30 + A \sin 30) = m(8.66 + 0.665) = 9.325 \cdot m \text{ N}$.

Q9 — (C) $A \tan(\theta)$

Geometric constraint: $a_{\text{vertical_block}} = a_{\text{horizontal_wedge}} \tan(\theta)$.

Q10 — (C) $f = \sqrt{k \cos^2(\theta) / m} / (2\pi)$

Effective spring constant along slope = $k \cos^2(\theta)$. $f = (1/2\pi) \sqrt{k_{\text{eff}} / m}$.

SECTION 6: ADVANCED PROBLEMS AND JEE TIPS

Problem Type 1: Wedge on smooth floor, block on smooth wedge, both released.

Method: Use momentum conservation (horizontal) + energy conservation.

This avoids differential equations entirely.

Steps: (1) Express all velocities using momentum constraint. (2) Apply geometric constraint. (3) Substitute in energy equation. (4) Solve for the one unknown (usually wedge speed V).

Problem Type 2: Find normal force N between wedge and block.

Use $N = m(g\cos(\theta) + A\sin(\theta))$ where A is the wedge acceleration.

A is found from the formula: $A = m g \sin \theta \cos \theta / (M + m \sin^2 \theta)$.

Final result: $N = M m g \cos(\theta) / (M + m \sin^2(\theta))$.

Check: $N < mg\cos(\theta)$ (block is lighter on free wedge than on fixed wedge). Physical!

Problem Type 3: Block on accelerating wedge, find minimum μ for no sliding.

In wedge frame: $a_{\text{rel}} = g\sin(\theta) - A\cos(\theta) - \mu N/m$ (if sliding down).

For no sliding: $a_{\text{rel}} = 0$ and static friction provides required force.

Net force along slope in wedge frame: $mg\sin(\theta) - mA\cos(\theta) = f_s \leq \mu_s N$

$N = mg\cos(\theta) + mA\sin(\theta)$

Condition: $mg\sin - mA\cos \leq \mu_s (mg\cos + mA\sin)$

$\mu_s \geq (g\sin - A\cos)/(g\cos + A\sin) = (g\tan - A)/(g + A\tan)$ [after dividing by $\cos$]

Common JEE Mistakes to Avoid:

Mistake 1: Using $1/k_{\text{eff}} = 1/k_1 + 1/k_2$ for spring along slope (wrong; effective $k = k$ here).

Mistake 2: Confusing angle θ from horizontal vs from vertical in wedge formulas.

Mistake 3: Forgetting that pseudo force is horizontal in wedge frame, not along slope.

Mistake 4: Not separating FBDs for wedge and block — drawing combined diagrams leads to errors.

Mistake 5: Using $A = g\tan(\theta)$ for rough wedge (only valid for smooth wedge).

QUICK REFERENCE — WEDGE FORMULAS

$$A_{\text{wedge}} = mg\sin(\theta)\cos(\theta) / (M + m\sin^2(\theta))$$

$$a_{\text{block along slope}} = (M+m)g\sin(\theta) / (M + m\sin^2(\theta))$$

$$N = Mmg\cos(\theta) / (M + m\sin^2(\theta))$$

$$\text{Block stationary on smooth wedge} \Rightarrow A = g\tan(\theta)$$

$$\text{Vertical block: } a_{\text{block}} = a_{\text{wedge}}\tan(\theta)$$

$$\text{Horiz spring equil: } x = mg\tan(\theta)/k$$

$$\text{Slope spring equil: } x = mg\sin(\theta)/k$$

$$\text{Momentum: } MV + mv_x = 0 \text{ (released from rest, smooth floor)}$$